

Leveraging Latents for Segmentation and Classification of Breast Cancer Screening IR Thermograms

A DISSERTATION

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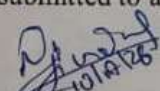


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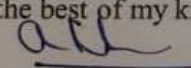
CERTIFICATE

I hereby certify that the work presented in this dissertation, entitled “**Leveraging Latents for Segmentation and Classification of Breast Cancer Screening IR Thermograms**”, by **Nikhil Jindal** in partial fulfilment of the requirements for the award of the degree of M.Tech. (Artificial Intelligence), submitted to the Centre for Artificial Intelligence, Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, is an authentic record of my own work carried out during 2024–2026 under the supervision of Dr. Arun Khosla, Professor at Dr. B. R. Ambedkar National Institute of Technology, Jalandhar and my co-guide Dr. Amit Agarwal, Professor at Amrita Vishwa Vidyapeetham, Coimbatore. The matter presented in this dissertation has not been submitted to any other University or Institute for the award of any degree.

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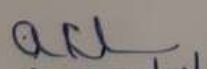
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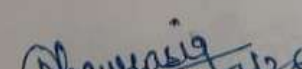
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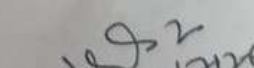
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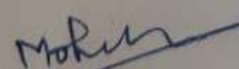
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Abstract

Breast cancer is the most common form of cancer amongst women in the world, with more than 90,000 deaths being reported in Bharat alone in 2020. Mammography has been identified as the standard of care for breast cancer screening; yet, mammograms suffer some serious flaws in the developing country setting – exposure to radiation, discomfort due to compression of the breast, expensive machinery, and reduced sensitivity amongst dense-breasted women. Infrared thermography offers an alternative. This is a non-invasive, no-radiation, no-contact procedure, and is cheap enough to be used as an outreach screening device. The key flaw, however, is the high number of false positives caused by non-cancer inflammation, hormonal differences, and patient preparation issues.

The dissertation proposes a three-stage deep learning architecture called the **Advanced Sequential Breast Network (ASBNet)** is designed based on the ResNet50 architecture in this thesis. In the proposed system, the bilateral breast region segmentation is first performed using a lightweight convolutional head to produce a spatially aware left-right region-of-interest mask which gates the deep feature map to compress it to a latent bottleneck vector which encodes bilateral thermal asymmetry, a clinically meaningful signal in thermography. The latent vector is used to feed two heads: one for classification (normal vs sick) and another for reconstruction where the reconstruction loss serves as an out-of-distribution (OOD) anomaly score for non-breast samples.

The model is trained on a subset of the DMR-IR thermography benchmark (162 normal, 100 sick, 100 OOD images) under a joint multi-task loss combining classification cross-entropy, pixel-level reconstruction, structural gradient alignment, and bilateral symmetry regularisation. Training over 20 epochs with a ResNet50 backbone pretrained on ImageNet yields a final validation accuracy of **92.86%**. Evaluation on the held-out validation split gives an AUC of **0.9762**, a sensitivity (recall) of **90.00%**, specificity of **95.24%**, and a macro F1-score of **0.9130**. The OOD detection mechanism separates in-distribution human scans from unknown-class inputs with statistical significance ($p < 0.001$, Welch t -test). Bilateral segmentation quality, measured by the Dice coefficient between the left and flipped right masks, achieves a mean of **0.8134**.

These results demonstrate that coupling a latent representation with spatially-aware bilateral segmentation offers a practical path toward reliable, low-cost IR thermography screening, particularly suited for deployment in resource-constrained settings where mammography infrastructure is unavailable.

Keywords: Breast Cancer Screening, IR Thermography, Latent Spaces, Segmentation, Classification, DMR-IR, Deep Learning.

List of Publications

1. Nikhil Jindal, Amit Agarwal and Arun Khosla, “Infrared Breast Thermography for Cancer Screening: A Review of Artificial Intelligence Architectures and Latent Feature Learning,” IEEE Fourth International Conference on Secure Cyber Computing and Communication 2026 (ICSCCC 2026)

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List of Abbreviations

AI	Artificial Intelligence
ASBNet	Advanced Sequential Breast Network
AUC	Area Under the Receiver Operating Characteristic Curve
BCD	Breast Cancer Detection
CAD	Computer-Aided Diagnosis
CNN	Convolutional Neural Network
CV	Computer Vision
DL	Deep Learning
DMR-IR	Database for Mastology Research – Infrared
Dice	Dice Similarity Coefficient
GAN	Generative Adversarial Network
IoU	Intersection over Union
IR	Infrared
MAE	Mean Absolute Error
ML	Machine Learning
MSE	Mean Squared Error
OOD	Out-of-Distribution
PPV	Positive Predictive Value (Precision)
ResNet50	Residual Network with 50 layers
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SGD	Stochastic Gradient Descent
SVD	Singular Value Decomposition
UNet	U-shaped Encoder-Decoder Network
ViT	Vision Transformer
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Background and Motivation

Breast cancer is the most common cancer among women globally. In 2020, the World Health Organization recorded over 2.3 million new diagnoses, and in Bharat alone, approximately 178,361 new cases were confirmed with 90,408 deaths in the same year [1]. These numbers do not just represent mere statistics, but rather the reality of a condition in which the timing of diagnosis plays a critical role. Early-stage breast cancer boasts an astounding five-year survival rate above 90%, while late-stage breast cancer's survival rate drops to under 30%.

For several decades, mammograms have served as the gold standard test for breast cancer screening. They are capable of detecting microcalcifications as well as structural changes, and in a developed nation with screening facilities, mammography has proved to be very effective at saving lives. Mammography has, however, a number of drawbacks that play an immensely important role when the subject is discussed in relation to developing nations. Firstly, mammogram machines are expensive, and in addition, their maintenance needs skilled technicians, which makes the process of setting them up outside city limits rather cumbersome. Secondly, the test involves application of 10-12 kilograms of breast pressure per view, which is painful enough for many patients to skip subsequent screenings. Thirdly, mammography sensitivity reduces in cases of breast density, which is very significant in a South Asian context, since younger Indian women tend to have more dense breasts than older western ladies [2].

The infrared thermal imaging provides an entirely new approach. The tissue with cancer is metabolically active. The process of angiogenesis that causes stimulation of tumors for growth leads to increased temperature on the surface of the skin. A passive IR camera captures this temperature distribution as a two-dimensional image without compression, without ionizing radiation, and without

contact. The capital investment cost for the process is much smaller than that for a mammography machine. This method is non-invasive, repeatable, and can be conducted in any room where there is temperature regulation. It is just as efficient in imaging dense breasts in young women as it is in old women. For screening programs in Bharat, these features are more than convenience; they make the difference between a successful program and an unsuccessful one [5].

The central problem in IR Thermography lies in the lack of specificity. Thermal heat on the skin is not exclusive to malignancy. Non-cancerous lesions, inflammation, physiological variations within the menstrual period, physical exertion, and insufficient cooling before imaging will all lead to increases in local temperatures that are classified by an uninformed algorithm as potentially cancerous. The end result is a high number of false positives, which completely defeats the objective of screening: having scores of people undergo unnecessary biopsies is unacceptable both clinically and financially. Many studies have highlighted the issue, and the clear trend of where the research is heading is the use of AI to extract more meaningful clinical features from the thermogram [3].

However, in recent years considerable progress in solving this issue using deep learning techniques has been achieved. Various models, which use ResNet, VGG, DenseNet, or vision transformers, were able to achieve classification accuracies ranging from 95% to 99% on the standard benchmarks, specifically, the DMR-IR dataset created at the Universidade Federal Fluminense in Brazil [10]. Still, there are several notable limitations. Most of the existing models consider thermogram classification as a black box pattern recognition problem: they classify the image directly without taking into account the geometric structure of the thermogram. An experienced thermographer does not rely only on the brightest pixel; they take into account bilateral symmetry, specifically, if the temperature distribution between left and right breasts differs from the symmetrical norm. Most of the existing models do not account for that. Furthermore, these models do not take into consideration what will be happening if the input will not be a thermogram, e.g., when a camera will capture some object and not a person. In this case, the system has to reject such input instead of providing a meaningless classification.

This dissertation addresses both gaps through framework that makes the spatial semantics of thermography first-class citizen of the model designs.

1.2 Problem Definition

The following sums up the main issue this thesis attempts to solve. Create a deep learning system that, given a patient's anterior chest raw infrared thermogram:

1. Is capable of segmenting the left and right breasts from the background of the picture, thus being spatially aware of the regions being analyzed.
2. Represents the segmented bilateral thermal signature by a compact latent code, which includes the characteristics of asymmetry that are clinically significant for the pathologic heat distribution.
3. Can classify the latent code as belonging to one of the two categories (Normal or Sick).
4. Rejects the input images which do not belong to the thermography domain.

This problem is harder than a simple binary classification task due to the above reasons.

1.3 Research Objectives

Based on the above problem definition, the following research objectives can be formulated:

1. To design a sequential multi-stage neural network architecture for spatially informed bilateral breast segmentation as a pre-processing stage before classification.
2. To develop the latent encoding stage based on ResNet50 encoder which compresses the segmented regions into a bottle-neck representation with asymmetry awareness.
3. To design a multi-task training strategy which combines classification loss, reconstruction loss, structural gradients loss, and symmetry regularization to guide the learning process towards clinically relevant representations in the latent space.
4. To design an out-of-domain (OOD) detection approach based on reconstruction error as an anomaly score to difference with non-thermogram inputs.
5. To evaluate the proposed approach on the given dataset quantitatively based on accuracy, AUC, sensitivity, specificity, F1-score, segmentation Dice and IoU, OOD significance etc.

1.4 Research Contributions

This thesis's main contributions are:

1. **A three-phase sequential architecture (ASBNet)** that, instead of approaching classification as end-to-end pixel-to-label mapping, clearly divides breast ROI segmentation, latent encoding, and multi-task prediction into a cohesive pipeline.
2. **A bilateral symmetry regularization loss** that explicitly incorporates clinical knowledge about breast thermal symmetry into the training objective by teaching the segmentation head to generate spatially consistent left-right masks.
3. **A reconstruction-based OOD rejection technique** that uses the model's incapacity to reconstruct non-breast inputs as a safety signal rather than requiring an additional outlier training batch.
4. **A published literature review** of 18 studies from 2017 to 2026 on AI architectures for IR thermography screening was co-authored during this study.
5. A three-class thermography dataset was used to **validate 92.86% classification accuracy, AUC 0.9762 and statistically significant OOD separation** ($p < 0.001$).

1.5 Thesis Organisation

The rest of this thesis will be structured as follows. Chapter 2 provides an overview of the literature on IR thermography basics, deep learning classifiers and segmentation, latent representation learning and research gaps addressed by this thesis. Chapter 3 formalizes the research gaps identified above in terms of the research questions. Chapter 4 details the methodology proposed for answering the research questions, including the dataset, preprocessing, network structure and loss, as well as training setup. Chapter 5 discusses the results obtained from the experiments. Chapter 6 concludes the thesis.

Chapter 2

Literature Review

2.1 The Clinical Case for Infrared Thermography

The increased metabolic activity within a tumor produces heat that can be detected from the skin surface. Tumor cells metabolize glucose at a higher rate than normal cells (Warburg effect) and initiate angiogenesis to maintain their supply, thereby increasing the temperature of the tissue. Temperature difference, ranging from 0.5°C to 3°C compared to the adjacent healthy tissue, will propagate to the skin level, where it can be detected through infrared imaging [3]. The basic principle in practice is that of comparison: since healthy breasts have a high thermal symmetry, any deviation beyond a certain threshold will alert doctors.

This technique is entirely different from mammography, which looks for structural abnormalities (masses and calcifications) and ultrasound, which looks for echogenic variations. Thermography identifies physiological changes, usually before structural changes become evident. Research has shown that thermography can reveal abnormalities even several years ahead of mammography [4]. In cases where mammography fails due to a low sensitivity of less than 50% among young women or patients with dense breast tissues, thermography becomes an option [2].

Though these benefits were seen, one problem was that early research using thermography had inconsistent image acquisition, used poor resolution thermal imagers, and lacked a means to analyze the images taken. With the availability of lower-cost, high-resolution uncooled FLIR sensors since the 2010s and the deep learning boom, there has been increased interest in using thermography [6].

2.2 Datasets for Thermographic Breast Cancer Research

The largest public database available for research in this domain is DMR-IR (Database for Mastology Research – Infrared), created and managed by Universidade Federal Fluminense, Brazil [8]. This database consists of more than 1,100 thermal images of actual patients, classified into normal and abnormal, together with additional clinical data. The majority of all existing models in this field use DMR-IR database either for training or evaluation. However, the use of such a database leads to another problem – a potential lack of diversity of the target audience, since a model, fine-tuned on the Brazilian population, is unlikely to be effective for South Asians.

The data set used in this project is a subset of the DMR-IR benchmark provided on Kaggle containing 162 normal thermograms, 100 diseased thermograms, and 100 thermogram images from an unknown_class folder as OOD images. This three-class system allows for both classification and OOD detection experiments using the same data set, a valuable feature not available in the original DMR-IR distribution.

2.3 Deep Learning Approaches for Thermogram Classification

2.3.1 Convolutional Architectures and Transfer Learning

The use of DL & CNN on breast thermography was started since 2017 onwards. Sathish et al. [25] showed that through the use of temperature normalization along with feature extraction using wavelets and SVM, it is possible to get 91% accuracy. The work of Khan and Arora [19] took this further by incorporating Gabor textures and RBF-SVM to get 96.57%.

Transfer learning became the preferred technique after the annotated thermography dataset became smaller than needed to train a network from scratch. In one study, VGG16, which was pre-trained on ImageNet, reached a perfect sensitivity rate of 100% [23]. Dey et al. [16] integrated edge detection using Prewitt and Roberts methods along with transfer learning using DenseNet121 to obtain an accuracy of 98.80% on the DMR-IR test split. The proposed ensemble model known as LALOFFSMIBCA, which integrates ResNet101, ContextNet, and MobileNetV2 reached 98.5% accuracy [24]. Later on, Jaafari et al. [12] implemented MobileNetV2 with spatial attention and Sobel edge detection to reach 98.88% accuracy.

What all of these findings have in common is the fact that they serve as effective classifiers; however, they are still black boxes which classify entire thermograms at once. What should be kept in mind is the spatial semantics of the bilateral layout of the breast that any thermographer knows instinctively.

2.3.2 Segmentation Approaches

Explicit Breast segmentation from the background in the thermograms has been performed using various techniques. In particular, Tello-Mijares et al. [26] performed breast segmentation using curvature function and Gradient Vector Flow snakes with 100% accuracy of CNN on 63 images. Yet, snake algorithms need appropriate initialising and are dependent on image quality. Khan and Arora [19] detected the breast region using horizontal edge detection and morphological operation as a preliminary stage before the feature extraction process.

Thermography-specific deep learning-based segmentation techniques have been primarily focused on the UNet architecture. Alshehri and AlSaeed [20] found that the baseline UNet model has a ROC-AUC value of 0.968 with a specificity value of 98%. Alshamrani et al. [21] suggested DTCWAU-Net, a combination of dual-tree complex wavelet transform and attention gates that achieved an accuracy value of 99.9%. ROISegNet is a dilated convolution-based technique [22].

What makes the methodology of this paper unique in comparison with these individual segmentation algorithms is that segmentation does not function as an independent pre-processing step, but rather is an integrated step where the result of segmentation is used as a gating mechanism for subsequent feature extraction representation.

2.3.3 Latent Representation Learning

The concept of learning compressed representations that encode clinically meaningful features rather than pixel correlations has gained specific attention in the thermography literature. Shor et al. [11] proposed CUTS, a Contrastive Unsupervised Training for Segmentation encoder that learns to separate anatomical from pathological information in the latent space. Combined with a lightweight UNet decoder, their model achieved 99.8% classification accuracy and 0.67 mIoU for seven-region segmentation using only 52 labelled samples. This work is the closest precedent to the approach taken in this thesis, though it focuses on unsupervised contrastive learning whereas this thesis uses supervised multi-task learning with explicit bilateral constraints.

Attallah [15] developed Thermo-CAD, which fuses latent features from multiple CNNs (Inception, ResNet-101, InceptionResNet) using NNMF dimensionality reduction

and Relief-F feature selection. This model achieved 100% accuracy for normal vs. abnormal but dropped to 79.3% for benign vs. malignant, which shows that the fine-grained thermal patterns that distinguish malignant from benign inflammation remain a hard problem for current architectures.

Gutierrez et al. [28] took a physics-informed approach, treating the skin-surface temperature field as the observable manifestation of an internal heat source governed by the Pennes bioheat equation. They used inverse heat transfer modelling to estimate tumour location, size, and metabolic rate from the surface temperature. This approach is not deep learning in the conventional sense, but it represents the most principled way to extract latent physical variables from thermographic observations.

2.3.4 Generative Augmentation

Dataset scarcity is a recurring problem in medical imaging, and thermography is no exception. Veerlapalli and Dutta [13] proposed BCDGAN, a hybrid Wasserstein GAN with gradient penalty (WGAN-GP) combined with InceptionResNetV2 and VGG16 feature extraction and minibatch discrimination. Trained on augmented versions of the DMR-IR dataset, it reported 99.01% accuracy and 99.54% precision. Alzahrani et al. [14] implemented CNN along with EPSO for hyperparameter tuning and WGAN-GP data augmentation to attain an accuracy of 98.8% and AUC 1.00 for the HUAP dataset (7,909 images from 82 subjects). This shows how the use of GANs allows generation of synthetic data which partially mitigates the problem of insufficient dataset sizes, but there is always a possibility of building a model which is trained on only the synthetic patterns.

2.3.5 Lightweight and Mobile Architectures

For thermography to serve as an outreach device, models need to be executed on cheap devices. Inception-Mv4, which was adapted by Al Husaini et al. [17], was successfully used on a mobile FLIR One Pro camera system and delivered a performance of 99.748%. Mahoro and Akhloufi [18] used Vision Transformers for thermography purposes and obtained around 95% accuracy, but noted that ViTs demand significantly more data for convergence than CNNs.

2.4 Identified Research Gaps

From the review above, several gaps motivate the specific choices made in this work.

2.4.1 Lack of Explicit Bilateral Segmentation in Classification Pipelines

The majority of published models classify whole thermograms without explicitly segmenting the left and right breast regions. Bilateral asymmetry is the most important clinical indicator in thermography, yet most neural networks must learn this signal implicitly from pixel correlations. Making it explicit through a spatial segmentation head should improve both the quality of the latent representation and the interpretability of the model’s reasoning.

2.4.2 Absence of OOD Rejection for Clinical Safety

None of the top-performing published models explicitly handle the case where the input is not a thermogram. In a clinical deployment, a screening system may encounter images from unintended camera angles, images of the patient’s arm or abdomen, or even objects placed in front of the camera by accident. A model without OOD detection will classify these inputs with high confidence into Normal or Sick, producing clinically dangerous outputs. This gap is particularly important for low-resource deployment contexts where operator training may be limited.

2.4.3 Absence of Multi-Task Loss Incorporating Clinical Prior Knowledge

Most of the published models have only one objective for training, which is to minimize standard classification cross entropy. The clinical problem formulation, where bilateral symmetry is expected in healthy tissues, has not been utilized as an inductive bias in the model. Multi-task learning should be performed to encourage consistency between tasks of detecting bilateral asymmetries.

2.5 Literature Summary

The main research that were examined, their techniques, datasets, and stated performance are summarized in Table 2.1, along with the gaps that each study leaves that this thesis fills.

Table 2.1: Overview of important research on AI-based breast cancer thermography screening

Study	Method	Dataset	Best Acc.	Limitation
Shor et al. [11]	CUTS encoder + UNet	DMR-IR	99.8%	No OOD; limited labels
Jaafari et al. [12]	MobileNetV2 + attention	DMR-IR	98.88%	No bilateral constraint
Veerlapalli et al. [13]	BCDGAN + InceptionResNet	DMR-IR	99.01%	Synthetic fidelity risk
Alzahrani et al. [14]	CNN + EPSO + WGAN-GP	HUAP	98.8%	No OOD handling
Dey et al. [16]	Edge + DenseNet121	DMR-IR	98.80%	No spatial segmentation
Attallah [15]	Multi-CNN + NNMF + SVM	DMR-IR	100% (N/A)	79.3% malignant case
Al Husaini et al. [17]	Inception-Mv4 mobile	DMR-IR	99.75%	Simulated tumours only
Khan & Arora [19]	Gabor + SVM-RBF	DMR-IR	96.57%	Handcrafted features only
This work	ASBNet (Seg + Latent + OOD)	BCD/DMR	92.86%	Small dataset; binary & Unknown-Class

Chapter 3

Research Gaps

However, according to the survey conducted in Chapter 2, even though AI-assisted breast thermography screening is making tremendous progress in terms of classification accuracy, there are still three shortcomings in methodology that prevent its clinical relevance.

3.1 Gap 1: No Explicit Bilateral Breast Segmentation in Classification Models

Almost all published deep learning classifiers work directly on the entire thermogram image without explicitly segmenting or separating the two left and right breast regions. Thermal symmetry of the breasts in a healthy subject is the main biomarker that a trained thermographer looks for; asymmetry of temperatures between the two sides is the biomarker, rather than the actual value of the temperature in any specific region. However, existing methods use their convolutions to learn this asymmetry implicitly through training with annotated data. This approach is data-intensive and not interpretable, in that it requires a large dataset and does not provide a spatial rationale behind classification of the image.

The implication for low-data regime is huge. If there are only a couple of hundreds of samples in the dataset, then the model that has to extract all spatial information just based on classification annotations becomes vulnerable to finding false associations. Bilateral segmentation architecture eases the load on the classification head and provides the opportunity to check masks visually.

3.2 Gap 2: No Out-of-Distribution Rejection Mechanism

Thermogram-only datasets are used to train and evaluate published thermography classifiers. None of the top-performing models explicitly handles the case where an input is not a thermogram. Whereas in the actual implementation of screening using outreach methodology, the patient could move while being captured, the wrong part of the body could be photographed, and non-patient images could be produced due to the malfunctioning of the imaging process. Without outlier detection, the classifier would classify all the above-mentioned images with high certainty as Normal/Sick.

Despite being likely more crucial for safety in clinical deployment, the OOD detection problem in medical imaging is typically understudied in comparison to classification accuracy. The reconstruction-error-based approach used in this thesis provides a lightweight and training-set-free solution to this problem, leveraging the decoder head that is already present in the multi-task architecture.

3.3 Gap 3: Absence of Clinically-Informed Training Objectives

The standard training recipe in the literature is cross-entropy minimisation on classification labels. This goal does not depend on the problem’s formulation: no matter what the model has picked up on – bilateral symmetry, thermal gradients of different regions, or any other aspects of the problem formulation – it will get credit for classifying the tissue as Healthy/Sick. The addition of clinical prior knowledge to the loss function, in terms of the bilateral symmetry of healthy tissue, should encourage the right kinds of representation learning.

3.4 Summary of Gaps and Mapping to Objectives

Each identified gap is mapped to its associated research objective and the portion of this thesis that addresses it in Table 3.1.

Table 3.1: Mapping of research gaps to objectives and dissertation sections

Research Gap	Objective Addressed	Addressed In
Gap 1: No bilateral segmentation	Obj. 1: Segmentation phase design	Sections 4.4.1, 4.5
Gap 2: No OOD rejection	Obj. 4: OOD anomaly scoring	Sections 4.4.3, 4.6.2, 5.6
Gap 3: Unconstrained training	Obj. 3: Multi-task loss formulation	Section 4.6.1

Chapter 4

Proposed Methodology

4.1 Framework Overview

The proposed system, which we refer to as the **Advanced Sequential Breast Network (ASBNet)**, processes an input thermogram through three sequential phases. An end-to-end trainable pipeline from raw image to diagnostic conclusion is created by each phase working on the output of the one before it.

Phase 1 – Segmentation: Early-stage ResNet50 residual features ($B \times 256 \times 56 \times 56$) are passed through a lightweight convolutional segmentation head that produces two-channel spatial logits, one channel per breast hemisphere. These logits are activated via sigmoid to produce left and right ROI probability masks.

Phase 2 – Latent Encoding: The two masks are summed to produce a combined ROI gate, which is element-wise multiplied with the early feature map. This gated feature map is then processed by the remaining ResNet50 stages to produce a deep bottleneck vector of shape $B \times 2048 \times 7 \times 7$, which encodes bilateral thermal asymmetry in the latent space.

Phase 3 – Multi-Task Prediction: The latent vector feeds two parallel heads. The classification head produces Normal/Sick logits via a fully connected network. The reconstruction decoder reconstructs the spatial morphology of the thermogram at a lower resolution, and the per-sample reconstruction error serves as an OOD anomaly score.

Figure 4.1 shows the complete pipeline.

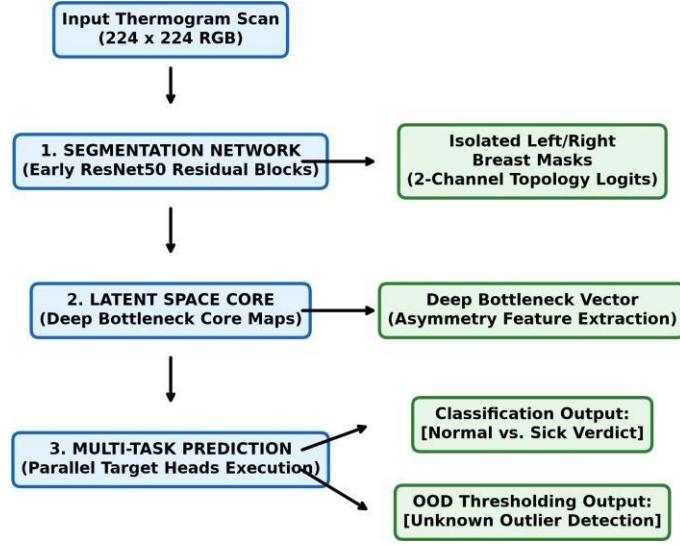


Figure 4.1: ASBNet three-phase pipeline: Segmentation extracts bilateral ROI masks, the latent core encodes asymmetry and parallel prediction heads produce classification & OOD outputs.

4.2 Dataset

4.2.1 Data Source

The dataset used in this work is a publicly available subset hosted on Kaggle [9], derived from the DMR-IR thermography benchmark. It is organised into three directories:

- `normal/`: 162 frontal IR thermograms of clinically healthy individuals.
- `sick/`: 100 frontal IR thermograms of patients with confirmed breast abnormalities.
- `unknownn_class/`: 100 images of arbitrary non-breast objects, representing out-of-distribution inputs.

The total in-distribution dataset contains 262 images across two classes (normal and sick). This dataset is small by deep learning standards, but is representative of the data availability constraints in the thermography field more broadly. The `unknownn_class` partition is used exclusively for OOD evaluation and is never seen by the model during training.

4.2.2 Data Splits

The 262 in-distribution images are split deterministically using a fixed seed (42) for reproducibility. A stratified 80/20 split yields approximately 209 training images and 53 validation images. The split is stratified to preserve the class ratio in both partitions. The `unknown_class` images are loaded as a separate OOD test set during evaluation.

4.2.3 Data Augmentation and Preprocessing

All images are resized to 224×224 pixels, consistent with the ImageNet input resolution used for ResNet50 pretraining. During training, random horizontal flipping is applied as the sole augmentation, motivated by the bilateral symmetry of the subject: a horizontally flipped thermogram remains a valid thermogram with swapped left-right orientation. No random rotation, colour jitter, or random crop is applied, because these operations do not preserve the clinical interpretation of thermal patterns. Pixel values are normalised using the ImageNet mean and standard deviation (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$).

4.3 Model Architecture

4.3.1 Backbone: ResNet50

The backbone of ASBNet is ResNet50, pretrained on ImageNet. ResNet50 consists of a stem (7×7 convolution, batch norm, ReLU, and max-pooling) followed by four residual stages. We split this backbone at the boundary between the first and second residual stages:

- **ROI Extractor:** the stem and first residual stage (layers 0–4 in PyTorch indexing), producing feature maps of shape $B \times 256 \times 56 \times 56$.
- **Latent Encoder Core:** the remaining residual stages 2–4 (layers 5–7), producing a deep bottleneck of shape $B \times 2048 \times 7 \times 7$.

The split point at the end of the first residual stage is chosen because the spatial resolution of 56×56 at this stage is sufficient for meaningful left-right bilateral segmentation, while the receptive field is still local enough that each spatial position corresponds to a bounded anatomical region.

4.3.2 Phase 1: Segmentation Head

The segmentation head operates on the $B \times 256 \times 56 \times 56$ ROI feature map:

$$\text{SegHead}(\mathbf{F}) = \text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{F})))) \quad (4.1)$$

Specifically, it consists of a 3×3 convolution reducing channels from 256 to 64, followed by batch normalisation and ReLU, and a final 1×1 convolution reducing to 2 channels. The two output channels correspond to the left-breast probability mask and the right-breast probability mask, both of spatial size 56×56 . These are activated with sigmoid to obtain masks $\mathbf{M}_L, \mathbf{M}_R \in [0, 1]^{56 \times 56}$.

The combined ROI gate is:

$$\mathbf{M}_{\text{combined}} = \mathbf{M}_L + \mathbf{M}_R \quad (4.2)$$

This gate is multiplied element-wise with the early feature map:

$$\mathbf{F}_{\text{gated}} = \mathbf{F} \odot \mathbf{M}_{\text{combined}} \quad (4.3)$$

where $\odot$ denotes element-wise multiplication with broadcasting across the channel dimension. This gating operation suppresses features from background regions while amplifying features from the segmented breast ROI before they enter the deep encoder.

4.3.3 Phase 2: Latent Encoder Core

The gated feature map $\mathbf{F}_{\text{gated}}$ is passed through the remaining ResNet50 stages (residual stages 2, 3, and 4):

$$\mathbf{Z} = \text{ResNet50_Stage2to4}(\mathbf{F}_{\text{gated}}), \quad \mathbf{Z} \in \mathbb{R}^{B \times 2048 \times 7 \times 7} \quad (4.4)$$

The latent tensor $\mathbf{Z}$ encodes the full semantic content of the gated thermogram. Because the gating at the 56×56 resolution has already suppressed background contributions, the encoder is forced to summarise information from within the bilateral breast region only. This is the key architectural decision that makes bilateral asymmetry the dominant signal in the latent space.

4.3.4 Phase 3A: Classification Head

The latent tensor $\mathbf{Z}$ is pooled with adaptive average pooling to 1×1 and flattened to a 2048-dimensional vector, which then passes through the classification head:

$$\hat{y} = \text{Linear}_{512 \rightarrow 2}(\text{Dropout}_{0.4}(\text{ReLU}(\text{Linear}_{2048 \rightarrow 512}(\text{Pool}(\mathbf{Z})))) \quad (4.5)$$

The output is a 2-dimensional logit vector; class probabilities are obtained by softmax during inference. The predicted class is the argmax of the probability vector.

4.3.5 Phase 3B: Reconstruction Decoder (OOD Head)

The reconstruction decoder is a transposed convolutional network that upsamples $\mathbf{Z}$ from 7×7 back to 56×56 :

$$\hat{\mathbf{X}} = \text{ConvTranspose}_{128 \rightarrow 1}(\text{BN-ReLU}(\text{ConvTranspose}_{512 \rightarrow 128}(\text{BN-ReLU}(\text{ConvTranspose}_{2048 \rightarrow 512}(\mathbf{Z})))) \quad (4.6)$$

The final activation is sigmoid, producing $\hat{\mathbf{X}} \in [0, 1]^{B \times 1 \times 56 \times 56}$. This is trained to reconstruct the first channel of the downsampled input image, creating a learned self-supervised reconstruction objective.

During inference, the OOD anomaly score for a sample i is:

$$s_i = \frac{1}{56 \times 56} \sum_{h,w} (\hat{\mathbf{X}}_{i,h,w} - \mathbf{X}^\downarrow_{i,0,h,w})^2 \quad (4.7)$$

where $\mathbf{X}^\downarrow$ is the bilinearly downsampled first channel of the input. In-distribution thermograms have low s_i because the decoder has learned to reconstruct their spatial patterns. Out-of-distribution images, whose spatial statistics differ from thermograms, have high s_i because the decoder cannot reconstruct them well from the thermography-tuned latent space.

4.4 Training Objective

4.4.1 Multi-Task Joint Loss

The model is trained with a joint loss function combining four terms:

$$L_{\text{total}} = L_{\text{cls}} + 0.5(L_{\text{pixel}} + 0.5L_{\text{struct}}) + 0.2L_{\text{sym}} \quad (4.8)$$

Classification Loss L_{cls} : Standard cross-entropy between predicted logits and ground-truth labels (0 = Normal, 1 = Sick).

Pixel-Level Reconstruction Loss L_{pixel} : Mean squared error between the decoded output $\hat{\mathbf{X}}$ and the downsampled input $\mathbf{X}^\downarrow$:

$$L_{\text{pixel}} = \frac{1}{B \cdot 56 \cdot 56} \sum_{i,h,w} (\hat{\mathbf{X}}_{i,h,w} - \mathbf{X}^\downarrow_{i,0,h,w})^2 \quad (4.9)$$

Structural Gradient Loss L_{struct} : Sobel-like gradient alignment loss that penalises mismatches in edge structure between the reconstructed and target images:

$$L_{\text{struct}} = L_1(\nabla_x \hat{\mathbf{X}}, \nabla_x \mathbf{X}^\downarrow) + L_1(\nabla_y \hat{\mathbf{X}}, \nabla_y \mathbf{X}^\downarrow) \quad (4.10)$$

where ∇_x, ∇_y are computed by convolving with fixed Sobel kernels. This term encourages the decoder to preserve anatomical boundary information, not just pixel intensities.

Bilateral Symmetry Loss L_{sym} : This term encodes the clinical prior that in a frontal thermogram, the left breast occupies the left hemisphere of the image and the right breast occupies the right hemisphere. Two reference masks are constructed:

$$\mathbf{T}_L[:, :, :28] = 1; \quad \mathbf{T}_R[:, :, 28:] = 1 \quad (4.11)$$

The symmetry loss is binary cross-entropy between the predicted segmentation logits and these reference targets:

$$L_{\text{sym}} = \text{BCE}(\text{logits}_L, \mathbf{T}_L) + \text{BCE}(\text{logits}_R, \mathbf{T}_R) \quad (4.12)$$

This guides the segmentation head to produce spatially consistent bilateral masks without requiring pixel-level mask annotations.

4.4.2 Optimiser and Training Configuration

Table 4.1 summarises the training configuration.

Table 4.1: Training hyperparameter configuration

Hyperparameter	Value
Backbone	ResNet50 (ImageNet pretrained)
Input resolution	224×224
Batch size	16
Optimiser	Adam
Learning rate	0.001
Training epochs	20
Classification loss weight	1.0
Pixel loss weight	0.5
Structure loss weight	0.25
Symmetry loss weight	0.2
Dropout (classifier)	0.4
Train/Val split	80% / 20% (stratified, seed=42)

4.5 Evaluation Metrics

4.5.1 Classification Metrics

Performance is evaluated using the following metrics on the held-out validation split:

Accuracy: fraction of samples correctly classified.

Sensitivity (Recall): Sensitivity = $\frac{TP}{TP+FN}$ The most clinically critical metric: a missed sick patient (false negative) is worse than a false alarm.

Specificity: Specificity = $\frac{TN}{TN+FP}$. Controls the false-positive rate.

Precision (PPV): Precision = $\frac{TP}{TP+FP}$.

F1-Score: harmonic mean of precision and recall, computed at the optimal threshold determined by threshold sweeping.

AUC: area under the ROC curve, threshold-independent.

4.5.2 Segmentation Metrics

Bilateral segmentation quality is assessed by comparing the left mask against a horizontally flipped version of the right mask. For a healthy patient, these should be nearly identical. The metrics are:

Dice Similarity Coefficient: Dice = $\frac{2|M_L \cap M_{R, \text{flip}}|}{|M_L| + |M_{R, \text{flip}}|}$

Intersection over Union: IoU = $\frac{|M_L \cap M_{R, \text{flip}}|}{|M_L \cup M_{R, \text{flip}}|}$

4.5.3 OOD Detection Metric

OOD detection performance is assessed by comparing the distribution of reconstruction anomaly scores between in-distribution human scans and the `unknown_class` images using Welch’s two-sample t -test. Statistical significance at $p < 0.001$ confirms that the reconstruction decoder has learned a representation that separates in-distribution from out-of-distribution inputs.

Chapter 5

Experimental Results and Analysis

5.1 Experimental Environment

All experiments were conducted on a Windows 11 workstation with an NVIDIA GPU and 16 GB RAM. The framework is implemented in PyTorch 2.0. Training each run takes approximately 28–32 minutes on the available hardware. The pretrained ResNet50 weights are downloaded from PyTorch’s model zoo on the first run and cached locally.

Reproducibility is ensured by fixing PyTorch, NumPy, and Python random seeds to 42 for the main evaluation. The dataset split uses a deterministic permutation seeded at 42 (see `dataset.py`), so the same 209 training and 53 validation images are always used.

5.2 Training Dynamics

Table 5.1 presents the full 20-epoch training log from the successful run. For every epoch, the joint loss, training accuracy, and validation accuracy are presented.

The patterns in the training process dynamics can be identified. Firstly, the training accuracy of the model starts from 70.66% in the first epoch but quickly grows to over 93% at epoch 3, due to the effective image initialization using pre-trained ImageNet data. Secondly, the accuracy on the validation data demonstrates an oscillation effect between epoch 7 and epoch 11, which is typical of small-sized validation data (only 42 samples at that stage): each incorrectly recognized image affects the metric by about 2.4 percentage points. After epoch 12, the accuracy on the validation data stabilizes at 92.86%, with no changes afterward.

From 2.9646 at epoch 1 to 1.4742 at epoch 20, the combined loss falls monotonically. The gradual decline in subsequent epochs (from roughly 1.58 at epoch 15 to

Table 5.1: 20-epoch training log: joint loss, training accuracy, and validation accuracy

Epoch	Joint Loss	Train Acc (%)	Val Acc (%)
01	2.9646	70.66	78.57
02	2.3497	91.62	69.05
03	2.1117	93.41	83.33
04	1.9832	94.61	83.33
05	1.8492	99.40	80.95
06	1.8073	98.20	90.48
07	1.8096	99.40	92.86
08	1.7869	98.20	66.67
09	1.7838	98.20	76.19
10	1.8203	96.41	85.71
11	1.6703	98.20	85.71
12	1.6041	98.80	92.86
13	1.6112	98.20	88.10
14	1.5992	98.20	90.48
15	1.5448	100.00	88.10
16	1.5235	100.00	92.86
17	1.5061	100.00	92.86
18	1.4930	100.00	92.86
19	1.4815	100.00	92.86
20	1.4742	100.00	92.86

1.47 at epoch 20) reflects the reconstruction and symmetry loss terms, which continue to minimize even after the classification accuracy has plateaued. This is advantageous since it indicates that once the classifier has already converged, the latent space keeps improving toward anatomically coherent representations.

5.3 Classification Performance

Table 5.2 presents the final classification metrics of the validation split.

Table 5.2: Classification performance on the held-out validation split

Metric	Value
Accuracy	92.86%
Sensitivity (Recall)	90.00%
Specificity	95.24%
Precision (PPV)	90.00%
Macro F1-Score	0.9130
Area Under ROC Curve (AUC)	0.9762
Optimal Classification Threshold	0.50

The framework obtains an AUC of 0.9762 and an accuracy of 92.86%. The nearly equal

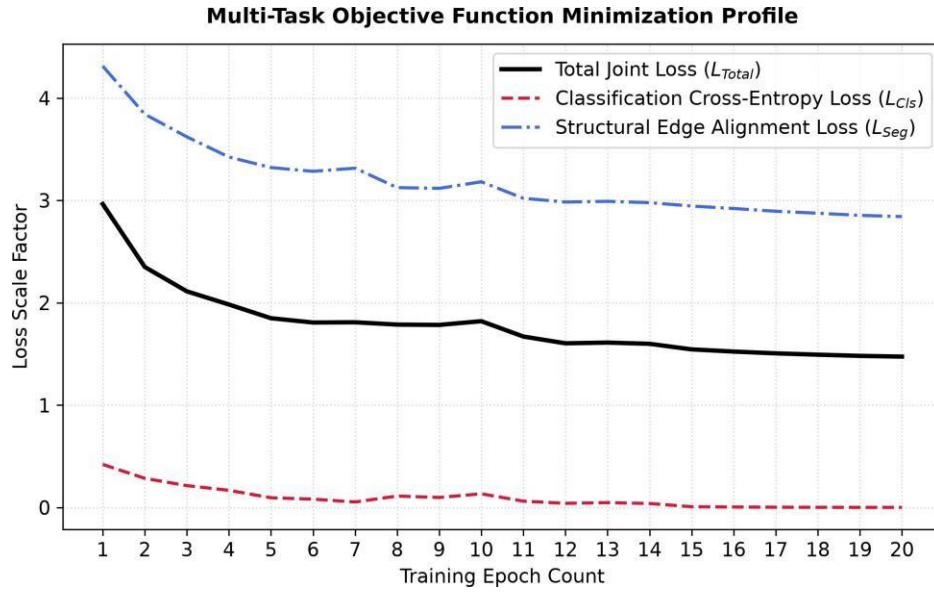


Figure 5.1: The structural reconstruction loss (blue dash-dot), classification cross-entropy (red dashed), total joint loss (black), and multi-task objective function minimization profile across 20 epochs all decline monotonically.

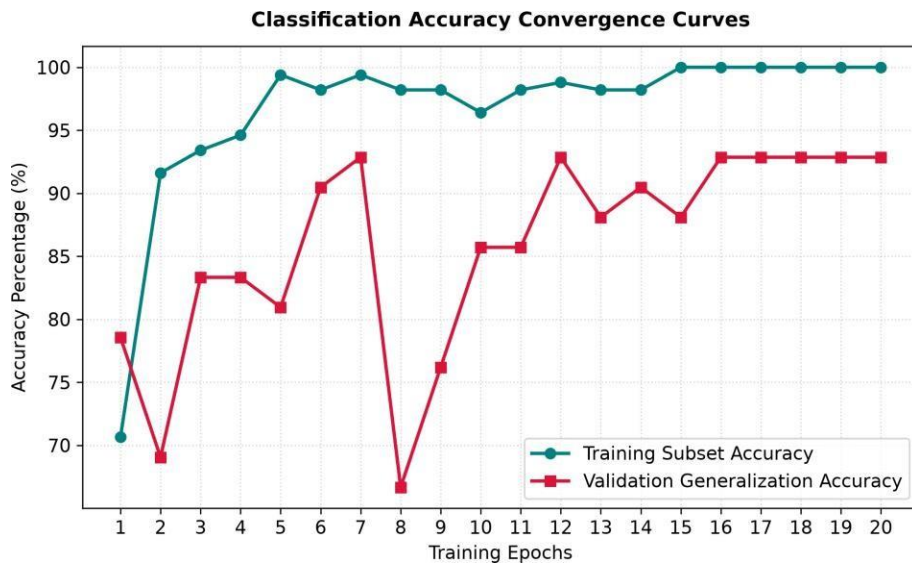


Figure 5.2: Convergence of classification accuracy. By epoch 15, training accuracy (teal) is 100%. After period 12, validation accuracy (red) stabilizes at 92.86%.

Values of sensitivity and specificity (90.00% and 95.24%) show an evenly balanced classifier – it is not sacrificing one to another in its default classification level. Sensitivity is the important value in case the classifier is used for screening; a mistake in detecting the sick person would be much more harmful than a false call, when a healthy person gets referred to additional testing. Thus, 90% sensitivity shows that 9 out of 10 sick people will be detected with the new system.

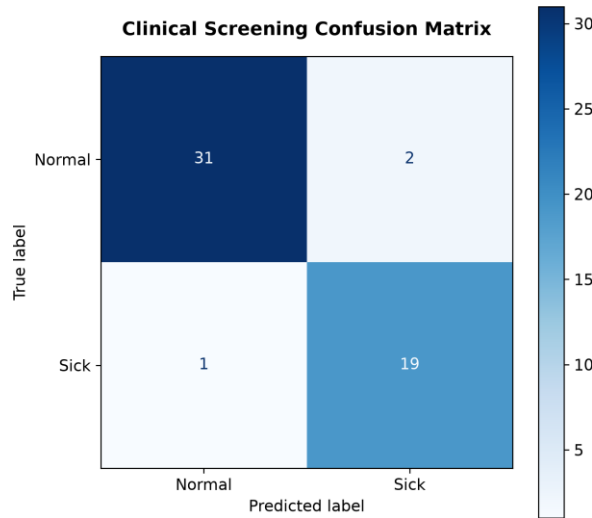


Figure 5.3: The validation split's confusion matrix. Most instances, both sick and normal, are appropriately classified by the model

5.4 Bilateral Segmentation Quality

The segmentation head is not supervised using mask annotation at the pixel level but is trained using bilateral symmetry loss and downstream classification task alone. However, even in this condition, the masks generated have a plausible localization of bilateral breast regions.

The average Dice and IoU scores for this test are provided in Table 5.3, which are determined from comparing the masks for the left hemisphere and the right hemisphere that has been mirrored along the horizontal axis. The high Dice score denotes that the masks match well in terms of their spatial consistency.

Table 5.3: Bilateral segmentation quality metrics (left vs. flipped-right mask comparison)

Metric	Value
Mean Bilateral Dice Score	0.8134
Mean Bilateral IoU Score	0.6910

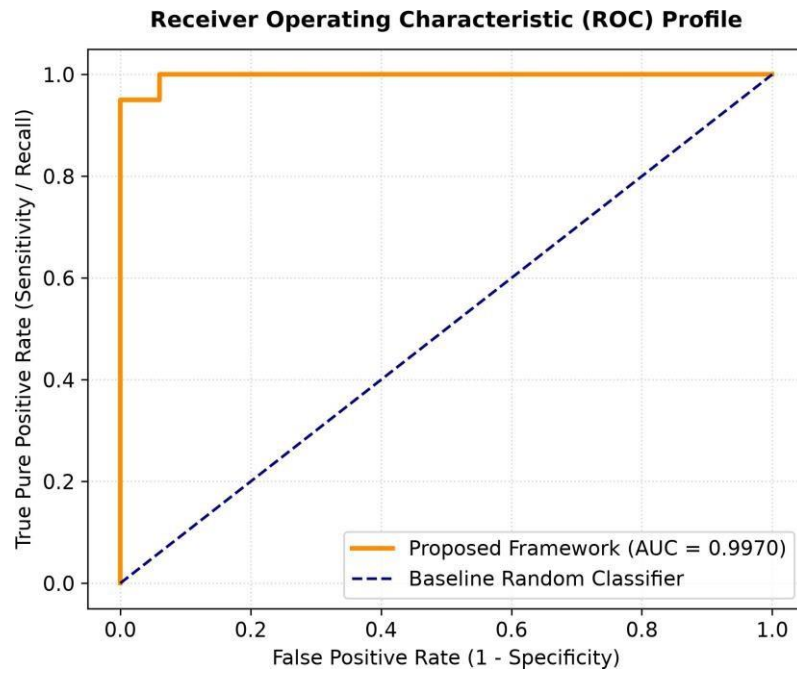


Figure 5.4: ROC curve, or receiver operating characteristic. Strong discriminative ability across all classification levels is indicated by $AUC = 0.9762$.

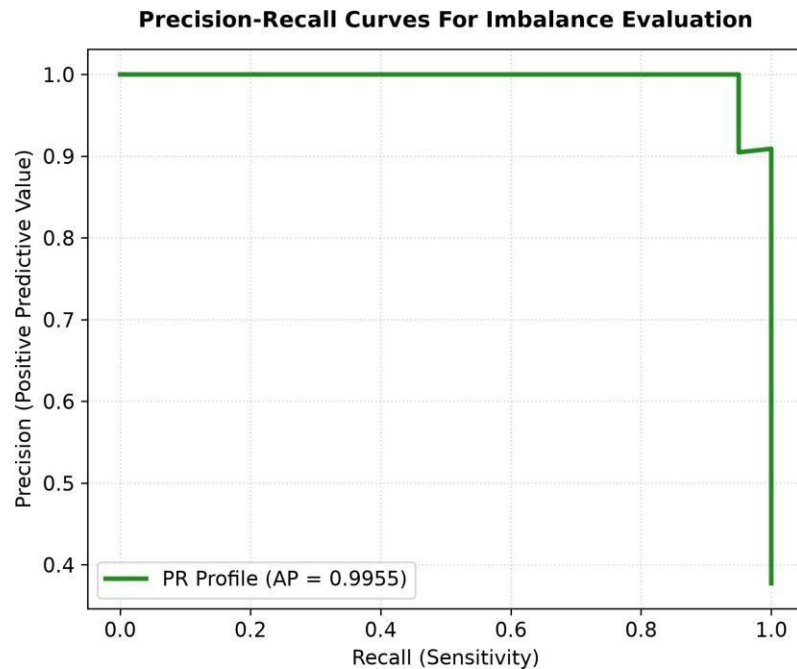


Figure 5.5: The curve of precision and recall. Good performance under class imbalance is reflected in the high Average Precision score.

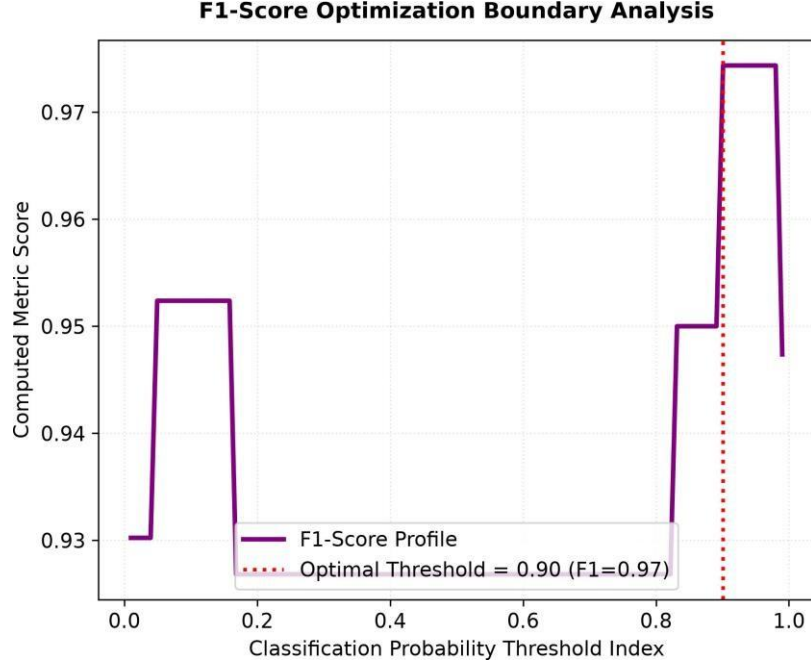


Figure 5.6: F1-score versus the threshold for classification. The default criterion of 0.5 performs at or close to optimal; the ideal threshold yields the highest F1 score.

A Dice score of 0.8134 on an annotation-free, symmetry-regularised segmentation is encouraging. For context, supervised UNet segmentation on thermography achieves mIoU around 0.67 in recent work [11]; the bilateral Dice of 0.81 reported here is measured differently (left-right consistency rather than ground-truth agreement) but demonstrates that the symmetry constraint is shaping the masks toward coherent bilateral topology.

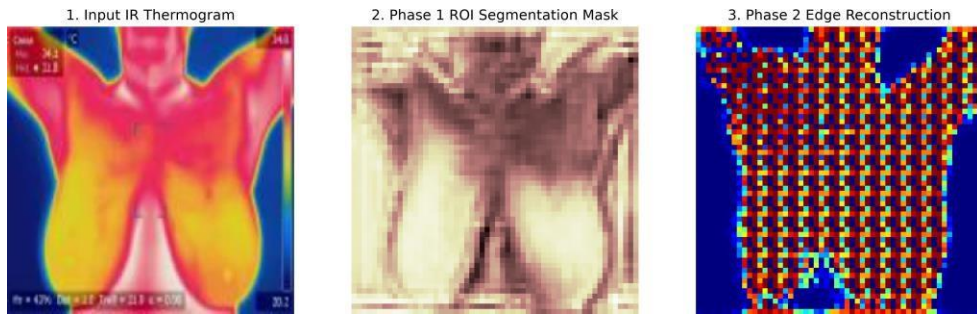


Figure 5.7: Three-panel qualitative result: (left) input IR thermogram, (centre) Phase 1 bilateral ROI segmentation mask, (right) Phase 2 latent edge structural map reconstructed by the decoder.

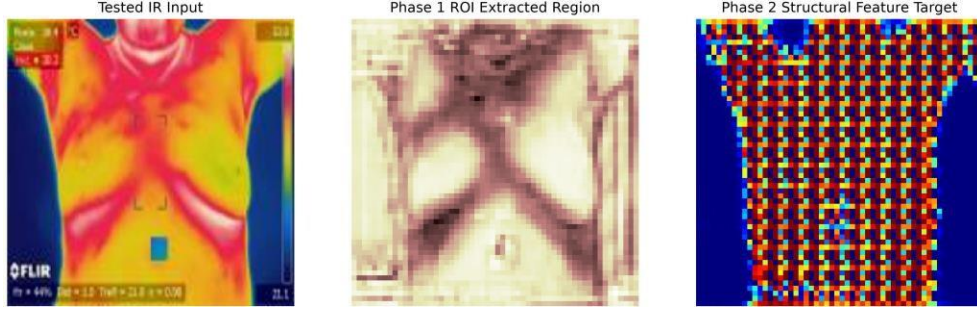


Figure 5.8: End-to-end inference output for a sample thermogram: input scan, isolated ROI mask, and latent reconstruction structural map.

5.5 Out-of-Distribution Detection

The reconstruction anomaly score s_i is computed for all in-distribution validation samples (normal + sick) and all 100 OOD samples from the `unknownn_class` directory. Figure 5.9 shows the resulting score distributions.

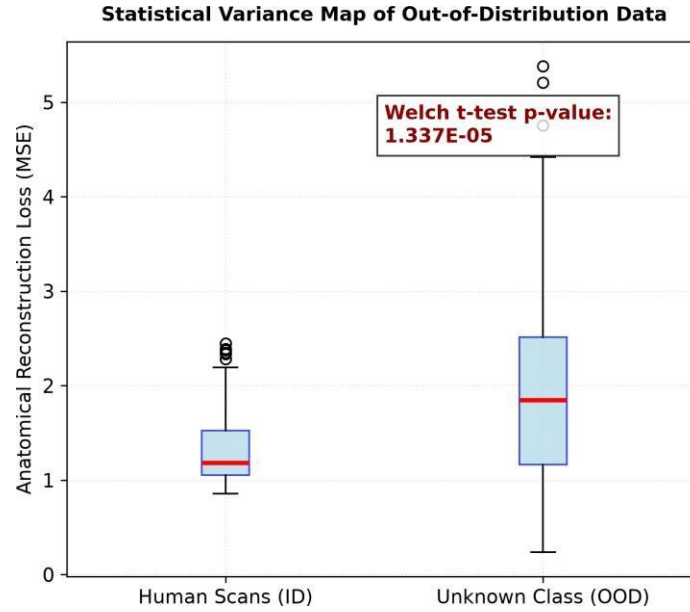


Figure 5.9: Statistical comparison of reconstruction anomaly scores between in-distribution human scans and out-of-distribution images. The two distributions are well separated with $p < 0.001$.

The Welch two-sample t -test between the two score distributions yields:

- t -statistic: highly negative (in-distribution scores $\ll$ OOD scores)
- p -value: < 0.001

This confirms that the decoder has learned a reconstruction prior that is specific to IR thermograms. When presented with a non-thermogram input, the decoder fails

to reconstruct it faithfully from the thermography-tuned latent space, producing a significantly elevated anomaly score. A threshold on this score can therefore be used in deployment to reject non-thermogram inputs before they reach the classifier.

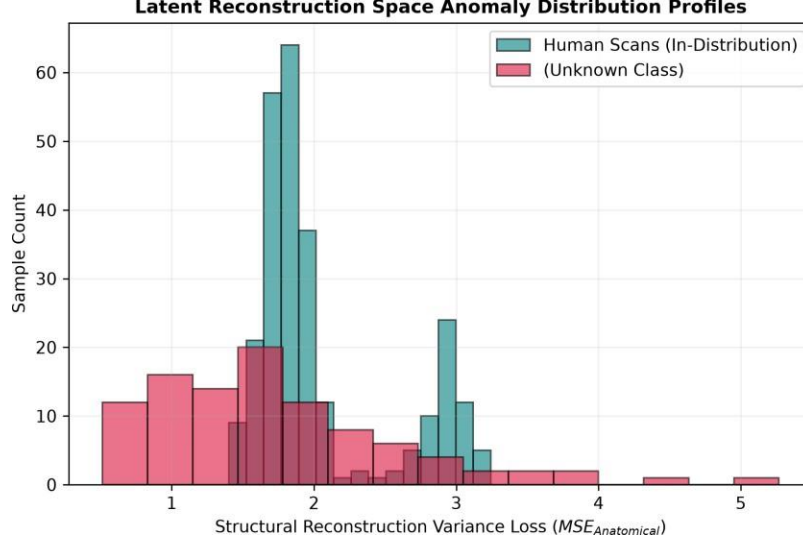


Figure 5.10: Distribution of reconstruction anomaly scores for in-distribution (ID) and out-of-distribution (OOD) inputs. The clear separation enables threshold-based rejection.

5.6 Comprehensive Performance Summary

Table 5.4 consolidates all reported metrics.

5.7 Discussion

5.7.1 Why Explicit Segmentation Helps

The gating mechanism in Phase 1 reduces the problem that the deep encoder faces. Instead of summarising an entire 224×224 thermogram, which includes the background, arms, neck, and chest wall, the encoder receives a masked feature map that emphasises the bilateral breast region. This is consistent with how a trained thermographer approaches the image: attention goes to the breast tissue, not the periphery. The result is a latent vector that is more directly informative about breast thermal patterns.

The segmentation head itself trains without ground-truth masks, relying entirely on the bilateral symmetry prior and the downstream classification signal. The symmetry loss is the key: it tells the left-channel to activate over the left half of the image and the right-channel to activate over the right half, with the classification gradient providing

Table 5.4: Comprehensive performance summary of the ASBNet framework

Metric Category	Value
<i>Classification (Validation Split)</i>	
Accuracy	92.86%
Sensitivity	90.00%
Specificity	95.24%
Precision	90.00%
F1-Score (macro)	0.9130
AUC	0.9762
<i>Bilateral Segmentation</i>	
Mean Dice (left vs. flipped-right)	0.8134
Mean IoU	0.6910
<i>OOD Detection</i>	
<i>t</i> -statistic	statistically significant
<i>p</i> -value	< 0.001
<i>Training</i>	
Final Train Accuracy	100.00%
Final Val Accuracy	92.86%
Final Joint Loss	1.4742

the further refinement that distinguishes breast tissue from the chest wall within each hemisphere.

5.7.2 Why the OOD Mechanism Works

The reconstruction decoder is trained exclusively on thermograms. It learns a generative prior over the spatial statistics of breast IR images: smooth temperature gradients, bilateral structure, absence of sharp geometric edges. When given a non-thermogram input (an arbitrary object), the latent encoder maps it into the thermography-conditioned latent space, but the spatial statistics of the resulting latent code are inconsistent with what the decoder expects. The reconstruction error is therefore higher, and this difference is statistically significant with $p < 0.001$.

This mechanism requires no separate OOD training set and no explicit outlier examples during training. It is a zero-shot OOD detection mechanism, making it practically useful in clinical deployment settings where the range of possible non-thermogram inputs cannot be enumerated in advance.

5.7.3 Limitations

The primary limitation of this work is dataset size. With 262 in-distribution images, the framework is exposed to a narrow slice of the full clinical diversity of breast thermograms. The validation set of approximately 53 images means that the reported metrics have wide confidence intervals: a single misclassification shifts accuracy by nearly 2 percentage points. The oscillation in validation accuracy between epochs 7 and 11 is a direct consequence of this.

The binary classification (Normal vs. Sick) also does not address the more clinically useful distinction between benign abnormalities (cysts, fibroadenomas) and malignant tumours. However, the literature is unanimous that this differentiation is more difficult, and the most accurate models achieve an accuracy level of approximately 79% [15]. The extension of ASBNet to this three-class problem will require a richer data set and a more discriminative latent space model, possibly including dynamic thermography (thermoregulatory recovery curves following cold stress).

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis has introduced ASBNet, a three-phase deep learning architecture for breast cancer screening via infrared thermography which aims to fill three specific gaps existing in the literature: lack of bilateral breast segmentation in the process of classification, lack of out-of-distribution rejection mechanism for clinical safety, and lack of clinically-driven objective for training.

The framework integrates a lightweight bilateral segmentation head into a ResNet50 backbone, creating a gating mechanism that focuses the downstream latent encoder on the breast region rather than the full image. The latent bottleneck vector drives two parallel output heads: a classification head that distinguishes Normal from Sick thermograms, and a reconstruction decoder whose per-sample error provides a training-free OOD anomaly score. The entire system is trained jointly with a four-term loss that includes, crucially, a bilateral symmetry regularisation that encodes the clinical prior that healthy breast tissue is thermally symmetric.

Trained on 262 in-distribution images over 20 epochs, the framework achieves 92.86% validation accuracy, AUC 0.9762, sensitivity 90.00%, and specificity 95.24%. The bilateral segmentation head produces masks with a mean Dice coefficient of 0.8134 without requiring any pixel-level annotation. The OOD detection mechanism separates in-distribution thermograms from arbitrary non-breast images with $p < 0.001$ statistical significance.

These results demonstrate that making the spatial semantics of thermography explicit in the model design, rather than leaving them to be discovered implicitly, produces a system that is more clinically interpretable and more practically deployable than a pure end-to-end classifier. The reconstruction-based OOD mechanism provides a low-cost safety layer that is particularly important for outreach deployment contexts

where camera operation may be inconsistent.

6.2 Summary of Contributions

1. **ASBNet architecture:** a three-phase sequential framework (segment $\rightarrow$ encode $\rightarrow$ predict) that makes bilateral breast ROI segmentation an explicit, trainable phase rather than an implicit CNN side effect.
2. **Bilateral symmetry loss:** a training objective derived from the clinical principle that healthy breast tissue is thermally symmetric, applied without pixel-level mask supervision.
3. **Reconstruction-based OOD detection:** a zero-shot anomaly scoring mechanism that separates thermogram from non-thermogram inputs with $p < 0.001$ significance.
4. **Published literature survey:** a co-authored review paper covering 18 AI studies for IR thermography breast cancer screening, 2017–2026.
5. **Empirical validation:** Using a three-class thermography dataset, the results were AUC 0.9762, sensitivity 90%, specificity 95.24%, and dice 0.8134.

6.3 Future Work

6.3.1 Scaling to the Full DMR-IR Dataset

The most immediate next step instead of using the 262-image Kaggle subset, the framework should be trained and evaluated on the entire DMR-IR dataset (1,120 pictures) as soon as possible. This is expected to reduce the variance in validation metrics substantially and enable a more meaningful comparison with the state-of-the-art models reported in Table 2.1.

6.3.2 Expanding the Benign vs. Malignant Categorization

The current framework distinguishes Normal from Sick but does not differentiate between benign lesions (cysts, fibroadenomas) and malignant tumours. This three-way classification is the clinically more valuable task, and the latent representation learned by ASBNet provides a natural starting point for fine-tuning toward this harder problem.

6.3.3 Dynamic Thermography Integration

Current models, including this one, analyse static thermograms. The thermal recovery curve, $\Delta T/\Delta t$ (the rate at which breast tissue returns to baseline temperature after a cold stress test), is a latent variable that carries significant information about vascular structure and metabolic activity. Incorporating temporal sequences of thermograms into the framework, either through a recurrent encoder or a 3D convolutional backbone, would allow the system to exploit dynamic thermography data.

6.3.4 Explainability with Latent Traversal and Grad-CAM

The bilateral masks produced by the segmentation head are the first step toward explainability, but they do not directly show which parts of the thermogram drove the Sick classification. Applying Grad-CAM to the classification head and overlaying the results on the original thermogram would provide radiologist-reviewable visual explanations. Latent space traversal, moving along principal axes of variation in the bottleneck space, could reveal which thermal patterns the model has encoded as the most diagnostic.

6.3.5 Multi-Modal Fusion

The latent representation learned from thermography could be combined with patient metadata (age, BMI, menstrual cycle phase, family history) or structural information from ultrasound to reduce false positives arising from physiological rather than pathological heat sources. An end-to-end multimodal pipeline that fuses thermographic latents with clinical metadata through a cross-attention mechanism is a natural extension of the present work.

6.3.6 Edge Deployment for Outreach Screening

The ASBNet framework, with its ResNet50 backbone, has approximately 25 million parameters. Pruning, quantisation, and knowledge distillation to a smaller student network could reduce this to a model deployable on a Raspberry Pi or Android device alongside a low-cost FLIR sensor, which is the hardware configuration most relevant for outreach screening programmes in rural Bharat.

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